

ETW3420-A1

33986010 A1 ETW3420

```
library(fpp2)
```

```
-- Attaching packages ----- fpp2 2.5.1 --
```

```
v ggplot2    4.0.2      v fma          2.5
v forecast   9.0.1      v expsmooth   2.3
```

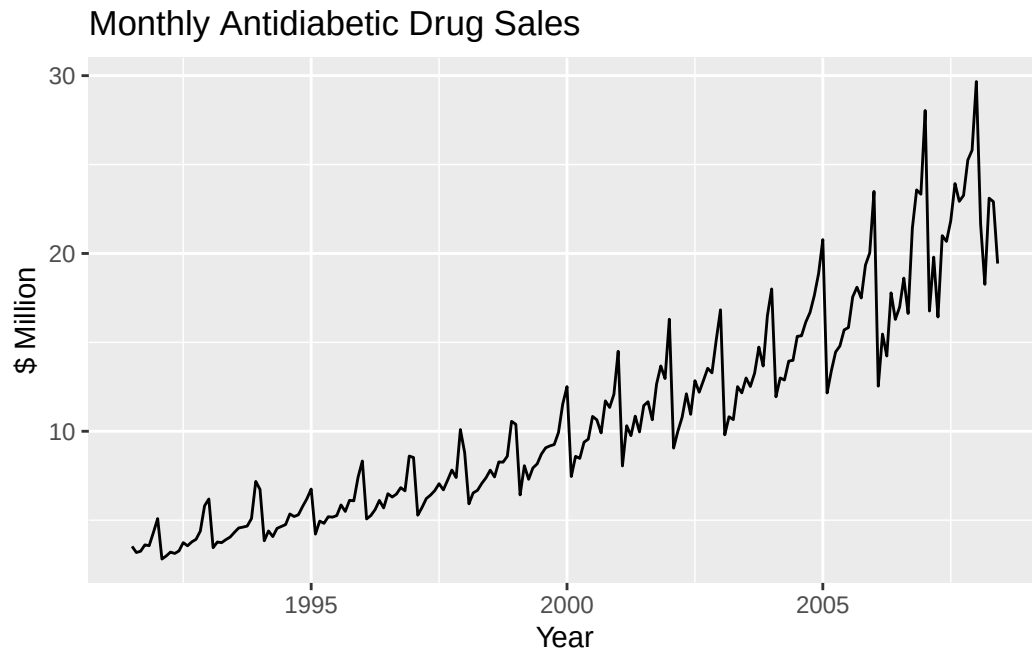
Question 1: Subseries Plots and Seasonal Analysis

```
help(a10)
```

```
starting httpd help server ... done
```

1. Time series plot

```
autoplot(a10) +
  ggtitle("Monthly Antidiabetic Drug Sales") +
  ylab("$ Million") + xlab("Year")
```

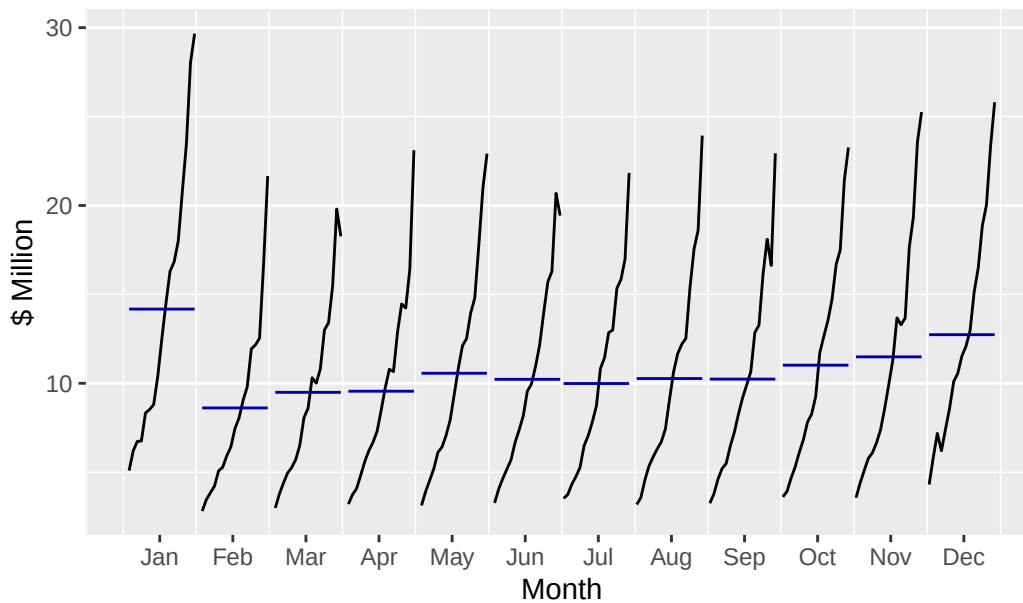


2. Seasonal subseries plot

- The grey lines represent the values for each month across different years.
- The blue horizontal line for each month indicates the average level for that month.

```
ggsubseriesplot(a10) +  
  ggtitle("Seasonal Subseries Plot of Anti-Diabetic Drug Sales") +  
  ylab("$ Million")
```

Seasonal Subseries Plot of Anti-Diabetic Drug Sales



3. Describe the seasonal pattern and comment on whether its magnitude changes over time.

[Time series plot]

There is consistent yearly seasonal pattern visible in the time series plot, because the ups and downs repeat in a regular way. In earlier years, the difference between peaks is relatively small. However, the fluctuations (ups and downs) become larger as the series increases, indicating growth strengthened in later years. The increase is accelerated in the year 2000, with higher peaks and higher average sales. Highest value are observed in the final year, showing continued expansion in sales.

The upward trend is supported by real-world evidence, the global anti diabetic drug market has been growing rapidly from around USD 97 billion in 2023 and is projected to reach nearly USD 300 billion by 2034, indicating continued expansion in demand (Future Market Insights, 2024). Furthermore, the increase of Type 2 diabetes worldwide is accelerated with a growing aging population, adoption of unhealthy lifestyles and poor dietary choices, alongside the introduction of new drug classes (e.g. GLP-1) (GlobeNewswire, 2025).

Australia is currently experiencing a diabetes epidemic with around 2 million Australians affected (diagnosed + undiagnosed). (Diabetes Australia, 2024). Overall, these factors explain the increasing trend and magnitude observed in the plot.

[Seasonal subseries plot]

In the seasonal sub series plot, it shows strong positive trend and strong seasonality where the monthly averages (blue lines) are differ noticeably across months. January records the highest

average, followed by a sharp decline in February, which have the lowest average. Sales then gradually increase throughout the remainder months of the year, suggesting an overall upward trend in addition to seasonality.

This seasonal pattern can be explained by institutional and behavioral factors within the Australian healthcare system. The peak in January is largely driven by the Pharmaceutical Benefits Scheme (PBS) Safety Net reset. Patients who reach their safety net threshold toward the end of the previous year often purchase medications in bulk during December or early January, when out-of-pocket costs are lower. The sales are not registered with the government until a week or two later. (Mellish et al., 2015)

The subsequent decline in February can be attributed to inventory saturation following this stockpiling behavior, as well as the shorter length of the month, which naturally leads to lower aggregate sales. Thereafter, the gradual rise as the year progresses is reflecting the continuous entry of new patients into the healthcare system and the increasing adoption of higher cost, modern anti-diabetic drug classes as the year progresses.

Question 2: Simple Forecasting Methods

```
help(ausbeer)
```

1. Apply mean, naive, seasonal naive method to training data

- Training period: 1992 Q1 to 2007 Q4
- Test period: 2008 Q1 onwards

```
# Train
beer_train = window(ausbeer, start = 1992, end = c(2007, 4))
# Test
beer_test = window(ausbeer, start = 2008)

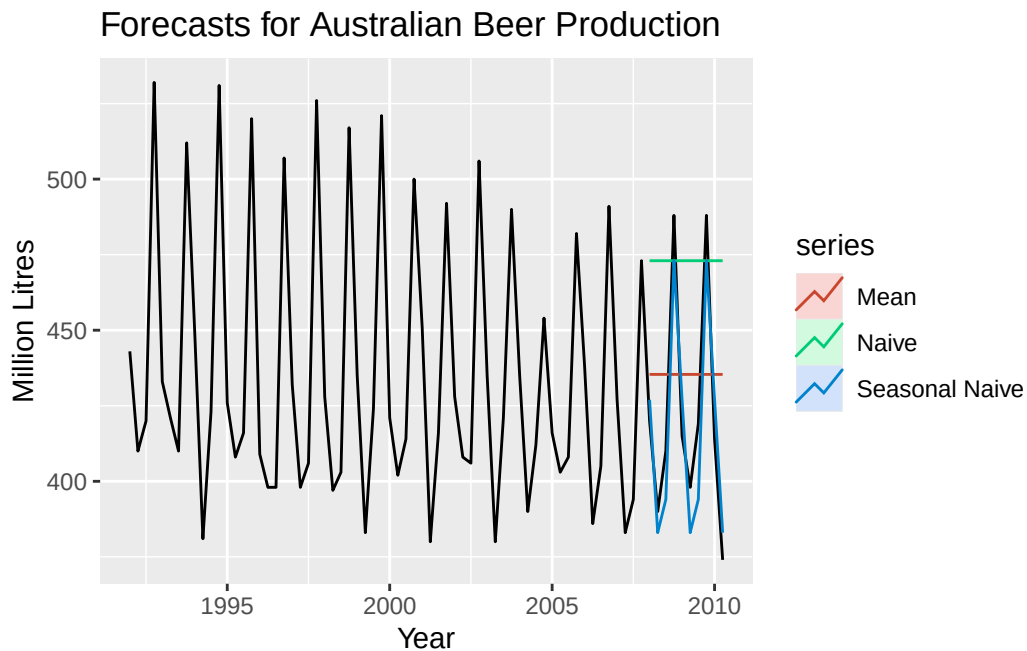
# No. of time periods in test set
h <- length(beer_test)
h
```

```
[1] 10
```

```
# Mean
mean_f <- meanf(beer_train, h = h)
# Naive
naive_f <- naive(beer_train, h = h)
# Seasonal Naive
snaive_f <- snaive(beer_train, h = h)
```

2. Plot forecast for the next 10 quarters using the 3 methods

```
autoplot(window(ausbeer, start = 1992)) +
  autolayer(mean_f, series = "Mean", PI = FALSE) +
  autolayer(naive_f, series = "Naive", PI = FALSE) +
  autolayer(snaive_f, series = "Seasonal Naive", PI = FALSE) +
  ggtitle("Forecasts for Australian Beer Production") +
  ylab("Million Litres") + xlab("Year")
```



3. Accuracy comparison of the three methods

```
beer_test <- window(ausbeer, start = 2008)

# Get accuracy results for each method
acc_mean <- accuracy(mean_f, beer_test)
```

```

acc_naive <- accuracy(naive_f, beer_test)
acc_snaive <- accuracy(snaive_f, beer_test)

# Combine into one table
acc_table <- rbind(
  Mean = acc_mean[2, ],      # Test set = row 2
  Naive = acc_naive[2, ],
  Seasonal_Naive = acc_snaive[2, ]
)

# View table
acc_table

```

	ME	RMSE	MAE	MPE	MAPE	MASE
Mean	-13.775	38.44724	34.825	-3.969866	8.283390	2.4353147
Naive	-51.400	62.69290	57.400	-12.954916	14.184424	4.0139860
Seasonal_Naive	5.200	14.31084	13.400	1.147554	3.168503	0.9370629

	ACF1	Theil's U
Mean	-0.06905715	0.801254
Naive	-0.06905715	1.254009
Seasonal_Naive	0.13184068	0.298728

```

# Better visualisation of table
library(knitr)

kable(acc_table, digits = 2, caption = "Forecast Accuracy Comparison (Test Set)")

```

Table 1: Forecast Accuracy Comparison (Test Set)

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1	Theil's U
Mean	-13.78	38.45	34.83	-3.97	8.28	2.44	-0.07	0.80
Naive	-51.40	62.69	57.40	-12.95	14.18	4.01	-0.07	1.25
Seasonal_Naive	5.20	14.31	13.40	1.15	3.17	0.94	0.13	0.30

To forecast the next 10 quarters of Australian beer production, the Seasonal Naive method is the most appropriate baseline, as it outperforms both the Mean and Naive methods. It achieves the lowest test-set errors, with an average deviation of approximately 14.31 megalitres (RMSE) and only about 3.17% errors (MAPE), indicating high predictive accuracy. Furthermore, its MASE value of 0.94 (less than 1) confirms that it performs better than a standard naive

benchmark, whereas the higher MASE values of the other methods indicate poorer forecasting performance.

The strong performance of the Seasonal Naive method is due to its ability to capture the underlying seasonal structure of the data by assuming that production in each quarter repeats the pattern of the same quarter in the previous year. In contrast, the naive method assumes that future values will equal the most recent observation, which ignores seasonal patterns and leads to poor performance. Similarly, the mean method forecasts using the overall average of the series, failing to capture both trend and seasonality.

As a result, the mean method and the naive method produce substantially larger errors (RMSE of 38.45 and 62.69, respectively) and noticeable bias, as reflected in their negative ME values (-13.78 and -51.40). This consistent underestimation highlights their inability to model recurring seasonal fluctuations, such as the post-holiday decline in production. Therefore, the Seasonal Naive method provides the most reliable forecasts among the simple approaches for this time

Question 3: Residual Diagnostics

```
help(goog200)
```

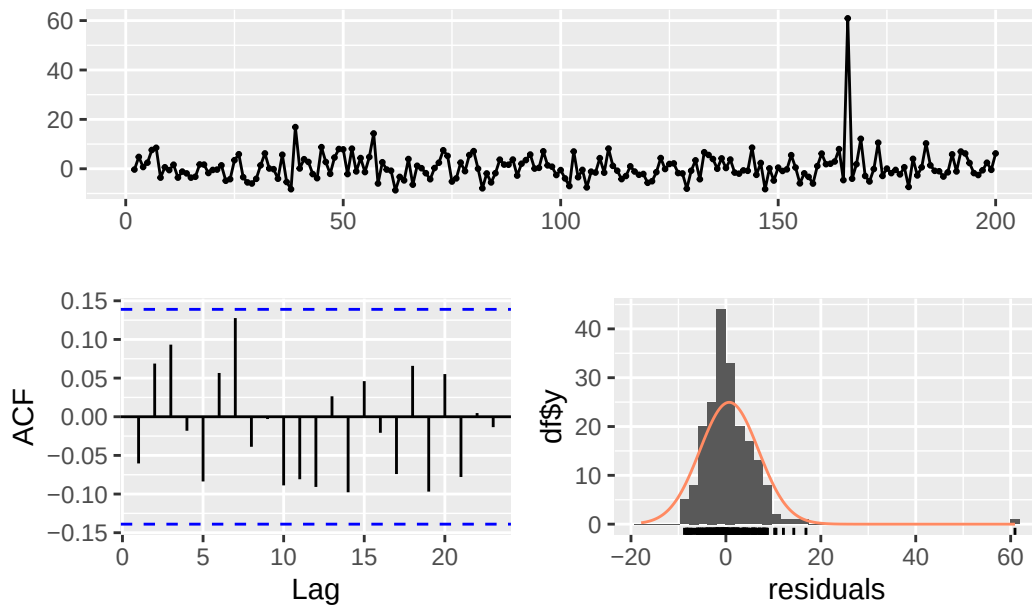
1. Fit naive model

```
fit <- naive(goog200)
```

2. Check residuals

```
checkresiduals(fit)
```

Residuals from Naive method



Ljung-Box test

```
data: Residuals from Naive method
Q* = 11.031, df = 10, p-value = 0.3551
```

```
Model df: 0.    Total lags used: 10
```

3. Interpret the ACF plot and Ljung-Box test results.

The output suggests that the Naive method is adequate because the residuals behave like white noise and does not exhibit any systematic patterns.

Time Plot (Top Panel)

The residuals fluctuate randomly around a mean of zero, which suggests that the Naive method is unbiased. However, there is a noticeable positive spike around $t = 165$. This corresponds to a specific day in July 2015 when Google's stock price increased significantly (approximately 16%) following an earnings announcement (BBC News, 2015). This outlier shows that while the model handles daily changes well, it cannot predict shocks or sudden market news.

ACF Plot (Bottom Left)

The ACF plot of residuals shows that most auto correlations lie within the 95% confidence bounds with no significant spikes, proving that there is no remaining autocorrelation in the

residuals. This suggest that the Naive method has effectively captured all the available patterns in the stock price movements. Hence, the daily change of closing stock prices behaves like white noise with no obvious trend, seasonality, or repeating patterns.

Histogram (Bottom Right)

The histogram shows that the residuals are roughly centered at zero and follow a bell-shaped curve, indicating residuals are normally distributed. However, it is slightly right skew caused by the July outlier which suggests that extreme movements may not be fully captured by standard prediction intervals.

Ljung-Box test

The Ljung-Box test results ($p = 0.355$) shows that the residuals from the Naive model are not autocorrelated. Since the p-value is greater than 0.05, we fail to reject the null hypothesis, indicating no strong evidence of autocorrelation left in the residuals up to lag 10. This confirms the model has captured most of the information in the data and residuals behave like white noise. Despite the presence of an outlier, there is no strong evidence of any remaining pattern such as trend or seasonality in the residuals.

4. State whether the residuals meet the assumptions of white noise.

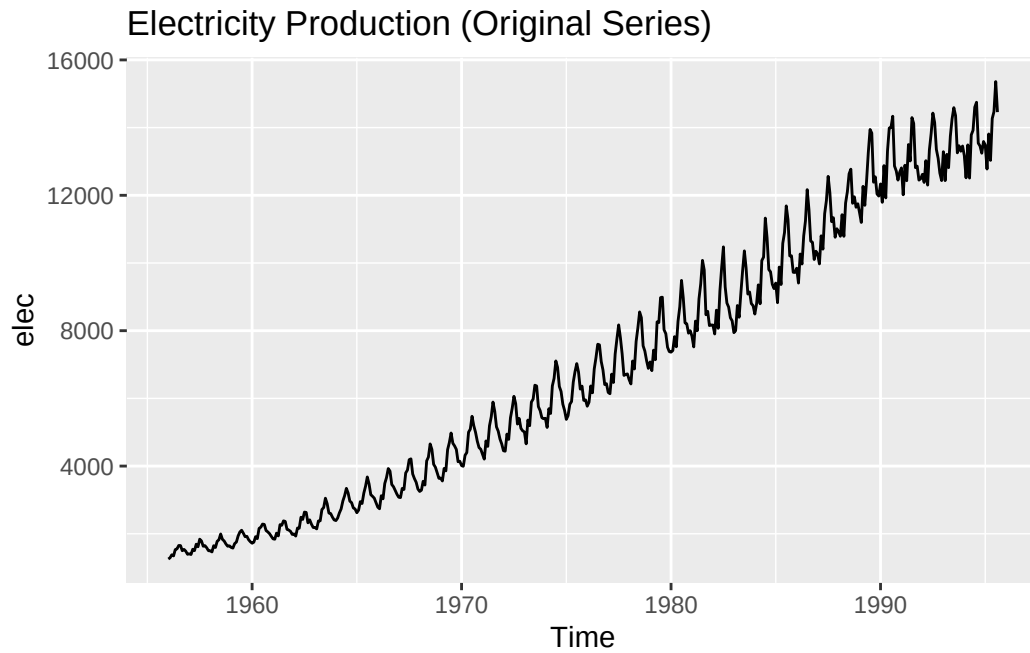
Yes, the residuals meet the assumptions of white noise, as they show no significant autocorrelation, are centered around zero, and have approximately constant variance over time. The ACF plot indicates that all autocorrelations lie within the 95% confidence bounds, and the Ljung-Box test ($p > 0.05$) confirms the absence of autocorrelation. Although there is a slight deviation from normality due to an outlier, the residuals overall behave like white noise.

Question 4: Box-Cox Transformation

```
help(elec)
```

1. Plot original series

```
autoplot(elec) +  
  ggtitle("Electricity Production (Original Series)")
```



2. Find lambda

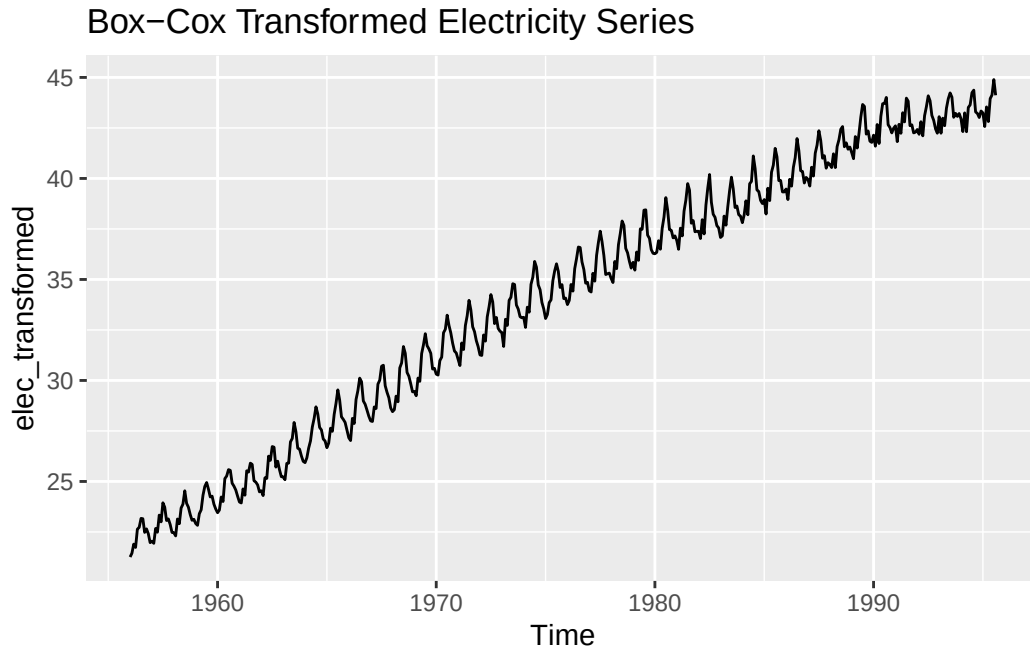
```
lambda <- BoxCox.lambda(elec)
lambda
```

```
[1] 0.2654076
```

3. Apply transformation and plot the transformed series

```
elec_transformed <- BoxCox(elec, lambda)

autoplot(elec_transformed) +
  ggtitle("Box-Cox Transformed Electricity Series")
```



4. Briefly explain why a transformation might be useful in time series forecasting.

The original elec dataset, representing Australian monthly electricity production exhibits increasing variability (multiplicative seasonality). This is evidenced by seasonal fluctuations that increase in magnitude proportionally as the overall trend rises from 1960 to mid-1990s. This pattern is driven by Australia's climatic extremes, general population and industrial capacity grow and seasonal differences in heating and cooling demand.

The estimated Box-Cox parameter is $\lambda = 0.2654$. Since this value is close to zero, it suggests the need of Box-Cox transformation to stabilize the variance. A lambda value between 0 and 1 means the transformation compresses larger values more than smaller values, resulting in a more even spread of the series over time. The transformation is similar to a logarithmic transformation.

Overall, applying a transformation stabilizes the variance and allows forecasting models to better capture the underlying trend and seasonal patterns in electricity production, thereby improving the accuracy and reliability of forecasts.

References

Ghosh, S. (2024, September 19). *Antidiabetics market outlook (2024–2034)*. Future Market Insights. <https://www.futuremarketinsights.com/reports/antidiabetics-market>

GlobeNewswire. (2025, February 20). *Diabetes drugs market trends, opportunities, and forecasts 2024–2030: Launch of novel drug classes like GLP-1 receptor agonists, SGLT2 inhibitors, and DPP-4 inhibitors transforming the landscape.* <https://www.globenewswire.com/news-release/2025/02/20/3029555/28124/en/Diabetes-Drugs-Market-Trends-Opportunity-and-Forecasts-2024-2030-Launch-of-Novel-Drug-Classess-like-GLP-1-Receptor-Agonists-SGLT2-Inhibitors-and-DPP-4-Inhibitors-Transforming-the-La.html>

Diabetes Australia. (2024). *State of the nation 2024: Diabetes in Australia.* <https://www.diabetesaustralia.com.au/wp-content/uploads/State-of-the-Nation-2024-Diabetes-Australia.pdf>

Mellish, L., Karanges, E. A., & Litchfield, M. J. (2015). *The Australian Pharmaceutical Benefits Scheme data collection: A practical guide for researchers.* BMC Research Notes, 8, 634. <https://pmc.ncbi.nlm.nih.gov/articles/PMC4630883/>

BBC News. (2015, July 17). *Google shares jump after strong earnings results.* <https://www.bbc.com/news/business-33572959>